

Important Chips on a Motherboard

Chip/Component	Description
CPU (Central Processing Unit)	The brain of the system. Performs all processing tasks.
BIOS/UEFI Chip	Firmware that initializes hardware at startup.
Chipset (PCH or Northbridge/Southbridge)	Controls communication between CPU, RAM, storage, etc.
VRM (Voltage Regulator Module) & MOSFETs	Regulate voltage to CPU, RAM, etc.
CMOS Battery & CMOS Chip	Stores BIOS settings, clock, etc.
RAM Slots & Memory Controller	RAM slots connect to CPU memory controller via chipset.
Super I/O Chip	Manages legacy ports (PS/2, serial), fan control, sensors.
Embedded Controller (EC) <i>(more common in laptops)</i>	Manages power, battery, keyboard, thermal controls.
Power Management IC (PMIC) <i>(mostly laptops)</i>	Controls power distribution to CPU, GPU, etc.
GPU (Graphics Processing Unit)	Renders graphics. Can be integrated (iGPU) or discrete (dGPU).
Clock Generator	Provides clock signals to sync system components.
Audio Chip (e.g., Realtek ALC)	Manages onboard audio.
LAN Chip (Ethernet Controller)	Handles wired networking.
Wi-Fi/Bluetooth Chips <i>(more common in laptops)</i>	Wireless connectivity chips.

Why It's Important for Troubleshooting

A non-functioning CPU means no POST or boot. Overheating, bent pins, or improper seating are common issues.
Corrupted BIOS can prevent boot. Symptoms: no display, beeping sounds. Can be reflashed or replaced.
Faulty chipset causes major system failures. Often not repairable; motherboard replacement needed.
Burnt or failed VRMs cause no power or instability. Check for short circuits or overheating.
Dead battery = loss of settings/time. Resetting BIOS (by removing battery) fixes boot/config issues.
Issues: bent pins, dirty slots, incompatible RAM. Beep codes indicate RAM failure.
If I/O devices or fan readings don't work, this chip may be faulty.
A faulty EC can prevent charging, power-up, or keyboard use.
Common failure point in laptops. Signs: won't charge or power on.
No display, artifacts, overheating = possible GPU failure. In laptops, dGPU failures are common.
If missing/failed, system won't boot or may hang.
No sound output/input; check drivers and hardware.
Ethernet not detected or slow connection can be due to this chip.
No wireless signal — check chip seating or antenna connection.